

Gangatharan I

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EDUCATION

Northeastern University, Boston, MA
Khoury College of Computer Sciences

Master of Science in Computer Science, GPA: 3.50/4.00

Related courses: Programming Design Paradigm, Algorithms, Database Management Systems, Foundations of AI

Sept. 2025 – Present
Expected graduation: May 2027

Vellore Institute of Technology, Chennai, India.

Bachelor of Computer Science and Engineering with Specialization in AI and ML, GPA: 3.4/4.0

Related courses: Object Oriented Programming, Data Structures and Algorithms, Databases, Software Engineering, Machine Learning, Deep Learning, Probability & Statistics

Sept 2021– April 2025

TECHNICAL KNOWLEDGE

Languages:	Java, Python, JavaScript
Frameworks:	Django, Spring Boot, React, PyTorch, NumPy, Pandas, Scikit-learn
Databases:	MySQL, MongoDB, PostgreSQL
Testing:	Junit5, Test-Driven Development
Tools & DevOps:	Docker, Git, GitHub Actions, CI/CD, REST APIs
Cloud:	Microsoft Azure (Microsoft Certified: Azure Administrator Associate)
Certifications:	Artificial Intelligence and Machine Learning Powered by Google Developers, Foundations of Software Engineering and Data Management Learning

WORK EXPERIENCE

Summer Research Internship, VIT Chennai CCPS Lab

Jun 2024 – Jul 2024

Research Intern

- Built a CNN–Transformer hybrid deep learning model for **healthcare diagnostics**, targeting breast cancer detection and multi-class abnormality classification, trained on **2,620 mammographic images** (CBIS-DDSM) with extensive image processing and **data augmentation** to reduce overfitting.
- Benchmarked and validated the model on the **MIAS dataset**, achieving **92.86% test accuracy** across 7 clinical classes, demonstrating strong cross-dataset generalization.
- Achieved **state-of-the-art performance** with **88.69% accuracy** and **0.95 AUC** on CBIS-DDSM; applied **SHAP-based explainability** to interpret model predictions and improve clinical transparency using **PyTorch, NumPy, and Matplotlib**.
- Work **published in Springer's** European Physical Journal Special Topics.

PROJECTS

Real Estate Management System

Sept 2025 – Dec 2025

- Designed and implemented **ETL workflows** using raw SQL via PyMySQL across a normalized relational schema (3NF, 10+ tables), with stored procedures, triggers, and user-defined functions automating data transformation and business logic.
- Designed **role-based access control (RBAC)** for Admin, Agent, and Client roles with session-based authentication (PBKDF2-SHA256 hashing, CSRF protection) and built complete **CRUD workflows** using raw SQL via PyMySQL - bypassing Django ORM for direct query-level control.
- Built a multi-filter property search engine with dynamic SQL query optimization across complex relational datasets, supporting filtering on price, location, type, and status; developed an analytics dashboard surfacing market KPIs, and deployed to production on Railway with a CI/CD pipeline via GitHub integration.

Calendar Application

Sept 2025 – Dec 2025

- Built a **modular Java scheduling platform** (MVC + Swing) supporting multi-calendar management, timezone handling, and complex recurring event logic across 5+ recurrence patterns.
- Designed **internal APIs** cleanly separating UI, business logic, and export layers (CSV, iCal), following SOLID principles and the Strategy + Factory design patterns.
- Achieved 85% + mutation coverage across **1,000+ JUnit5 tests** using PiTest, validating edge cases in timezone conversion and recurrence resolution.